DSA 210 Spring Project

Does a spike in job-related Google searches predict a rise in youth unemployment with a time delay, or is it simultaneous?

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1. Motivation

Youth unemployment in Turkey has remained high for over a decade, and official TUİK figures come out with a delay of several weeks. This project investigates how many months in advance job-related Google searches predict a rise in youth unemployment. When labor market conditions worsen, people tend to search for job listings and employment services before any official report captures the shift. The goal is to measure the size of this time delay and test whether it is statistically consistent.

2. Research Questions

Main question

Does a spike in job-related Google searches predict a rise in youth unemployment with a time delay, or is it simultaneous?

Sub-questions

- Is the correlation between search terms and unemployment strongest at 0, 1, or 2 months ahead?
- Which search terms show the strongest correlation with youth unemployment at different time delays?
- Did the relationship between search behavior and unemployment change during periods of economic shock, such as the 2018 currency crisis or COVID-19?
- Do search terms associated with lower-educated job seekers behave differently from those associated with university graduates?

3. Data Description

This project uses four data sources, all at monthly frequency and covering January 2015 to December 2024. All sources are merged into a single dataset using the month as a shared key, giving 120 observations and 16 variables in total.

Source	Variables	Observations
TUİK Labour Force Statistics	Youth unemployment rate (ages 15-24)	120 monthly
Google Trends	10 job-related search terms	120 monthly
TCMB EVDS	Inflation, USD/TRY, interest rate, industrial production	120 monthly
TCMB EVDS	Unemployment insurance applications	120 monthly

Search terms collected: işkur, iş ilani, iş arıyorum, linkedin, kariyer.net, iş kurumu, cv hazırlama, işsizlik maaşı, işsizlik sigortası, işsizlik ödeneği.

4. Methods

- EDA: time series plots, correlation matrix
- Cross-correlation analysis: Pearson correlation at time delays of 0 to 6 months
- Hypothesis testing: independent samples t-test *-high vs low search volume months-*
- Machine learning: Linear Regression and Random Forest, evaluated with 5-fold cross-validation

5. Results

5.1 Exploratory Data Analysis

The time series overview shows youth unemployment peaked around 2019 at roughly 27% and has been declining since. The COVID-19 period is visible as a sharp dip in industrial production in early 2020. Inflation and the USD/TRY exchange rate both increased sharply after 2021.

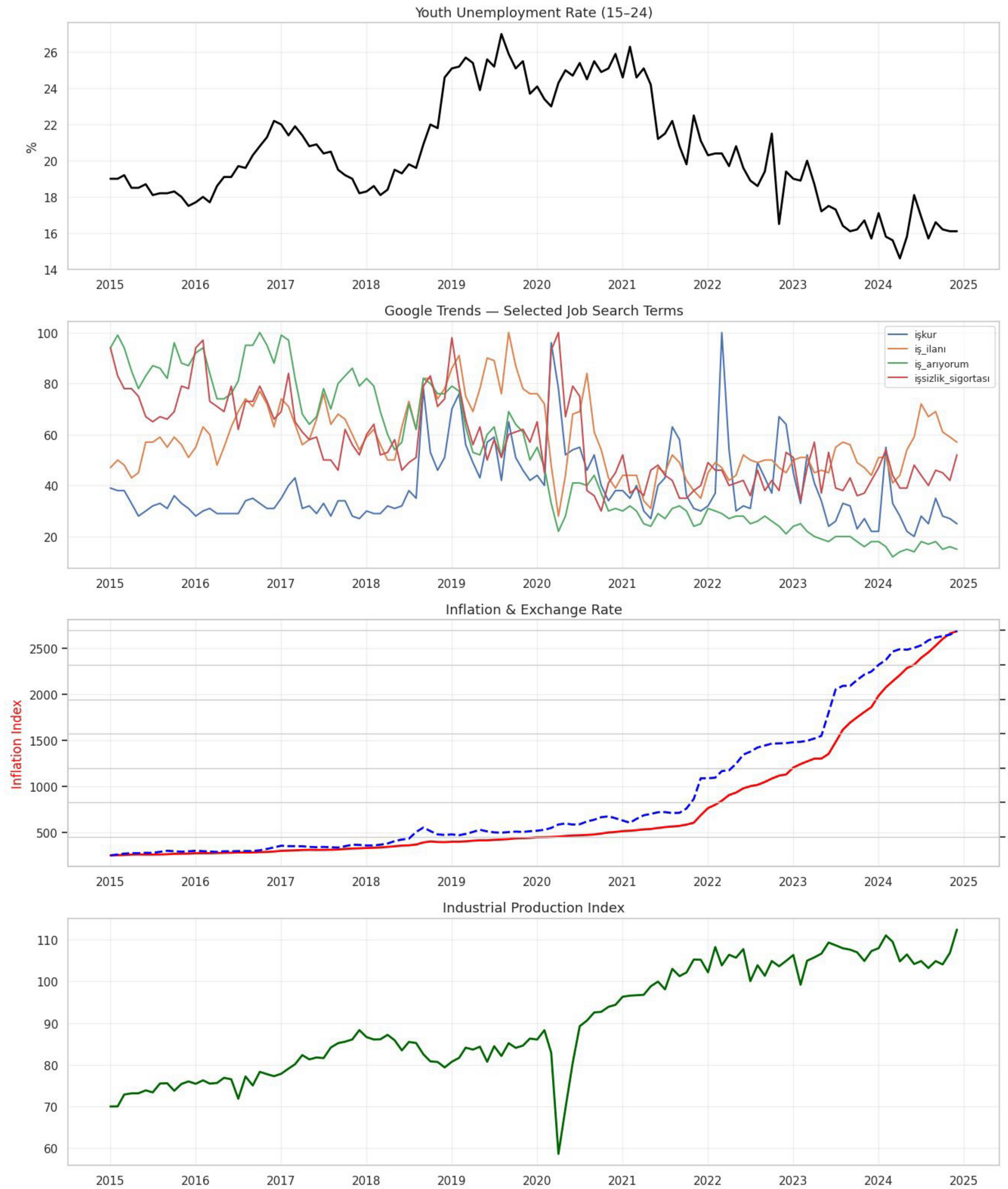

Figure 1. Time series overview: youth unemployment, Google Trends, inflation, exchange rate, and industrial production (2015-2024).

The correlation matrix shows that işkur (r =0.48) and iş_ilanı (r =0.33) have the strongest positive correlations with youth unemployment. Inflation (r = -0.55) and USD/TRY (r = 0.53) show negative correlations, which reflects that the recent high-inflation period coincided with a decline in youth unemployment.

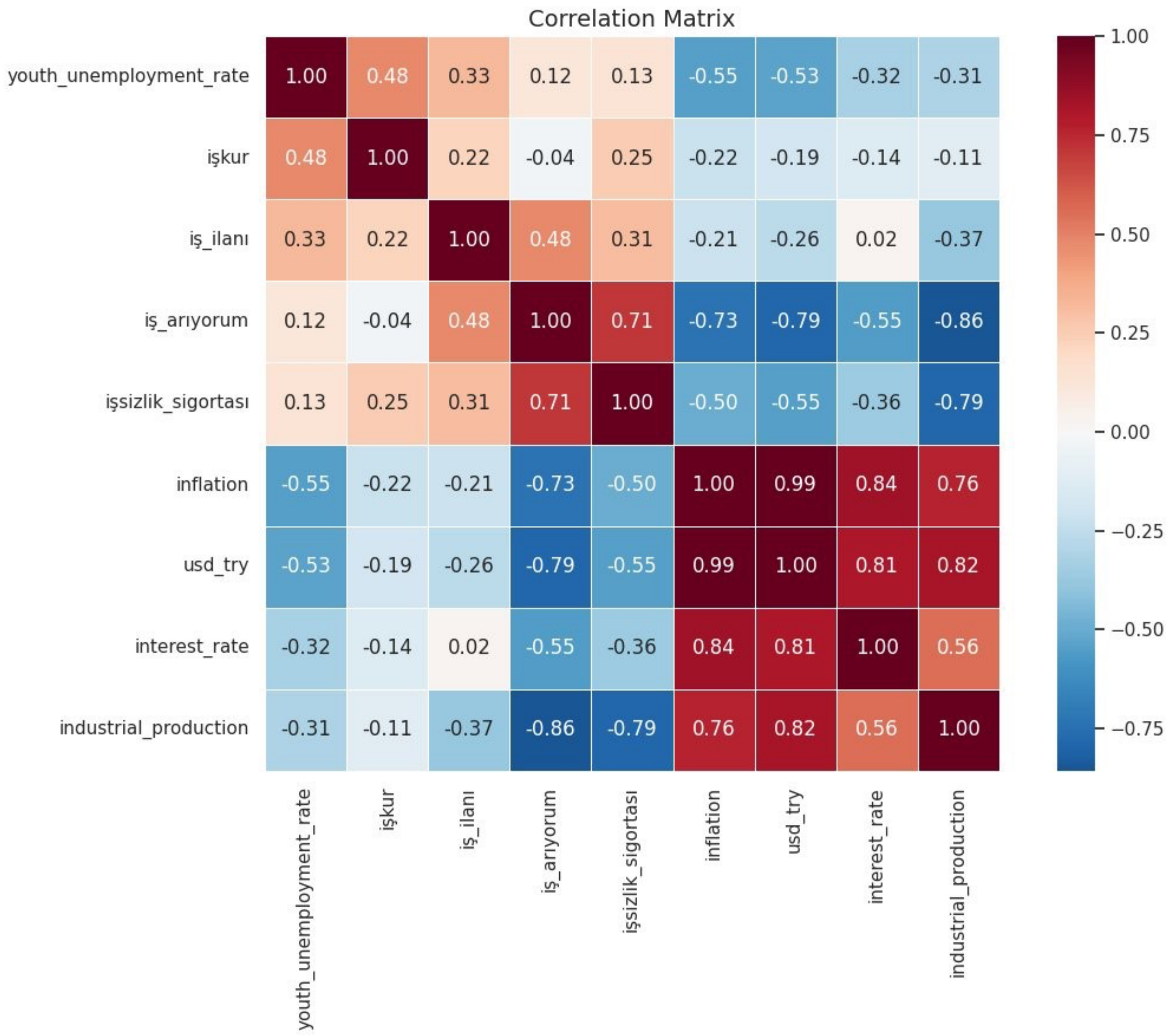

Figure 2. Correlation matrix across all variables.

5.2 Cross-Correlation Analysis

To test whether search data leads unemployment, We compute Pearson correlation between each search term and youth unemployment at time delays of 0 to 6 months. A growing correlation as the delay increases means the search term is moving ahead of the unemployment figures.

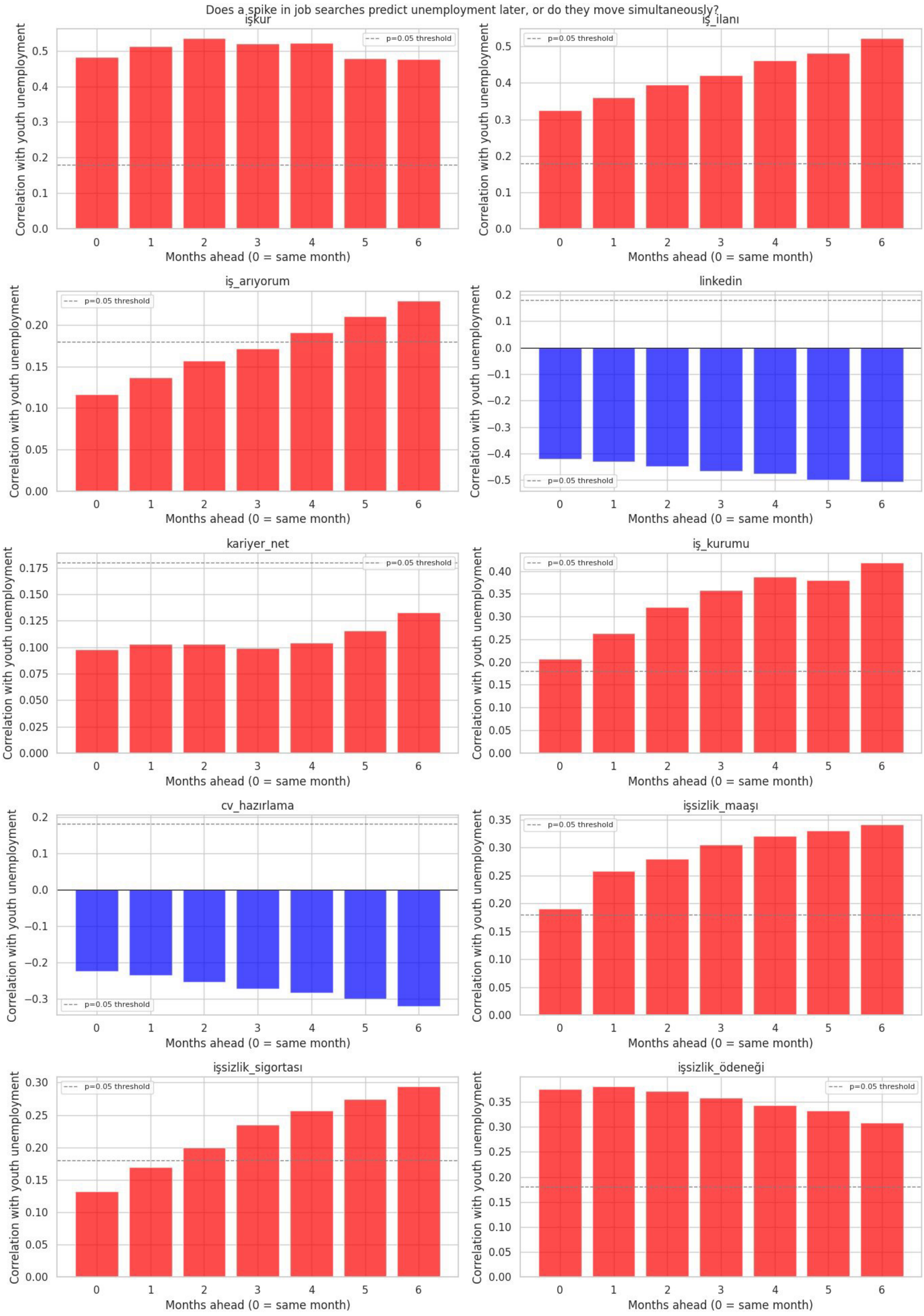

Figure 3. Cross-correlation between each search term and youth unemployment at time delays of 0 to 6 months.

Key findings:

- işkur shows the strongest and most consistent signal, peaking at lag 2 ($r = 0.54$, $p < 0.05$)
- iş_ilanı shows a growing correlation with lag, peaking at lag 3 ($r = 0.43$, $p < 0.05$)
- linkedin and cv_hazırlama show consistent negative correlations across all lags
- iş_arıyorum shows no statistically significant relationship at any lag
- işsizlik_ödeneği has a significant positive correlation at lag 0, suggesting it reflects real-time distress

5.3 Hypothesis Testing

We split months into two groups based on whether search volume was above or below the median, then tested whether youth unemployment 2 months later was significantly different.

H0: Mean youth unemployment is the same in high and low search volume months.

H1: Mean youth unemployment is higher in high search volume months.

Term	High mean	Low mean	t-stat	p-value	Result
işkur	22.0%	18.7%	6.90	< 0.001	Reject H0
iş_ilanı	21.0%	19.9%	2.04	0.044	Reject H0
iş_arıyorum	20.8%	20.1%	1.22	0.225	Fail to reject H0

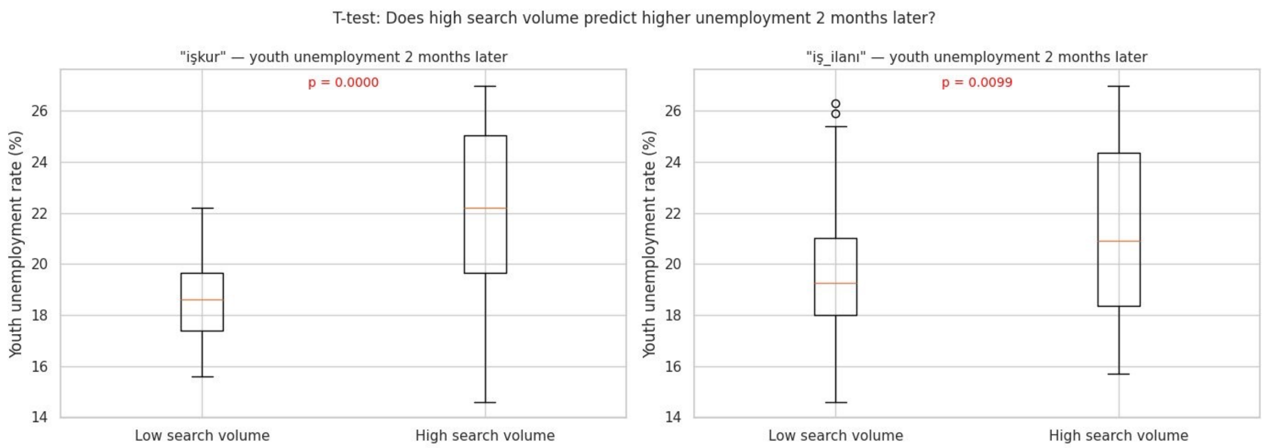

Figure 4. T-test: youth unemployment 2 months later in high vs low search volume months.

For işkur, months with high search volume are followed by unemployment averaging 22.0% two months later, compared to 18.7% in low-search months. This 3.3 percentage point gap is highly significant ($p < 0.001$).

5.4 Machine Learning

I trained two regression models to predict youth unemployment from search terms and macroeconomic variables. Search terms were lagged by 2 months. Models were evaluated with 5-fold cross-validation.

Model	CV RMSE	CV R2
Linear Regression	1.455	0.754
Random Forest	1.174	0.848

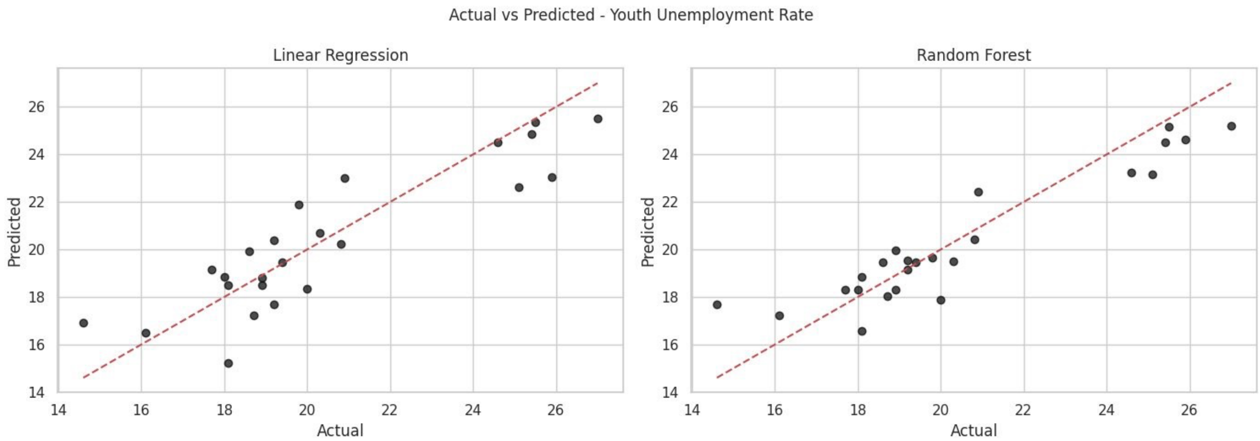

Figure 5. Actual vs predicted youth unemployment rate for both models.

Random Forest performed better on 5-fold cross-validation (R2 = 0.848 vs 0.754). Both models show consistent test and CV scores, with no signs of overfitting.

5.5 Feature Importance

At lag 2, işkur_lag2 is the single most important feature, accounting for roughly 28% of the model's predictive power. At lag 1, inflation becomes the top feature with işkur_lag1 close behind.

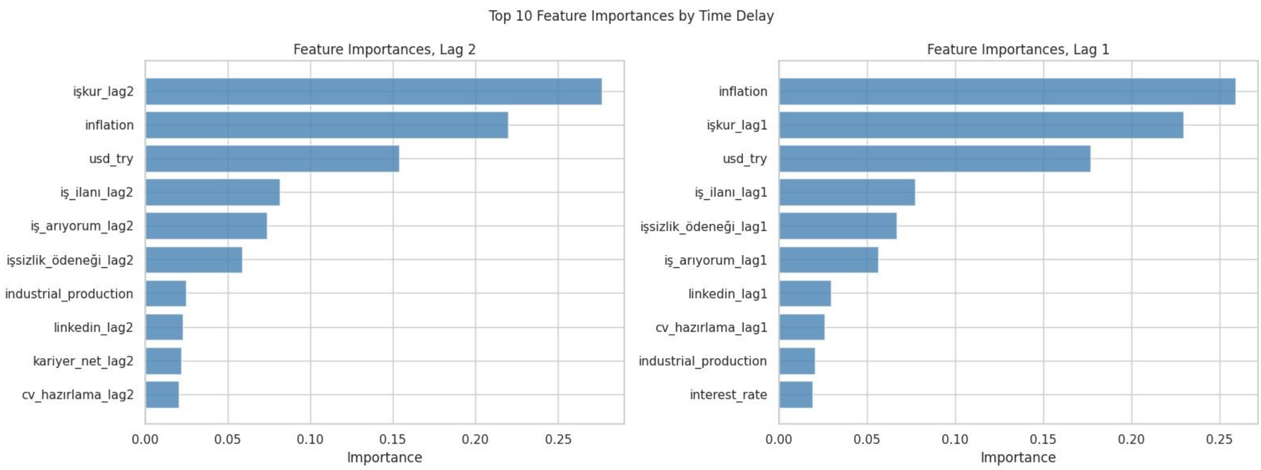

Figure 6. Top 10 feature importances for Random Forest at lag 1 and lag 2.

5.6 Residual Analysis

Residuals are roughly centered around zero with no strong directional pattern, suggesting the model does not systematically over- or under-estimate unemployment.

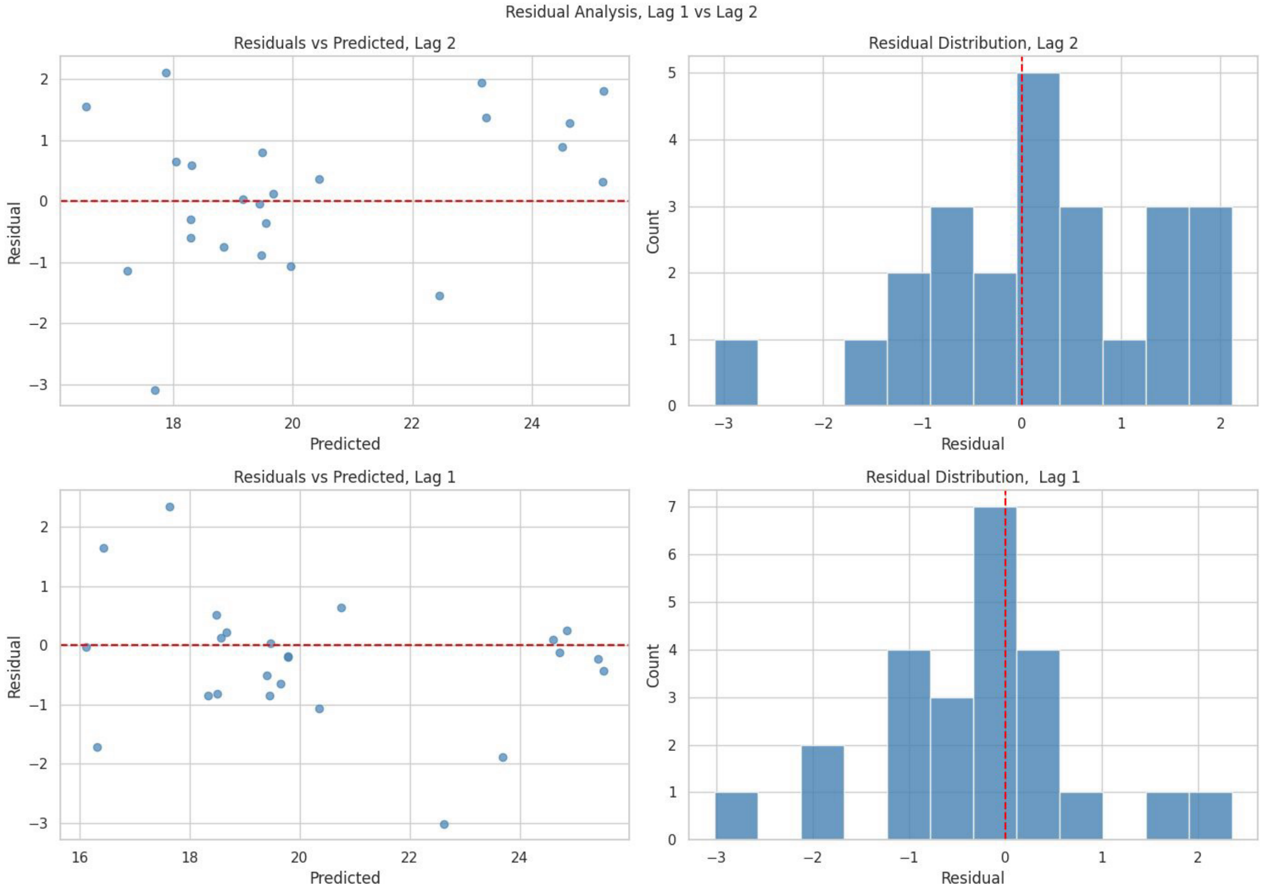

Figure 7. Residual analysis for lag 1 and lag 2 models.

5.7 Time Delay Comparison

I tested Random Forest performance at lags 0 through 3. Lag 1 produces the best R2 (0.874) and the lowest RMSE (1.121), slightly outperforming lag 0 (R2 = 0.863). This directly answers the research question: search data leads unemployment rather than moving with it simultaneously.

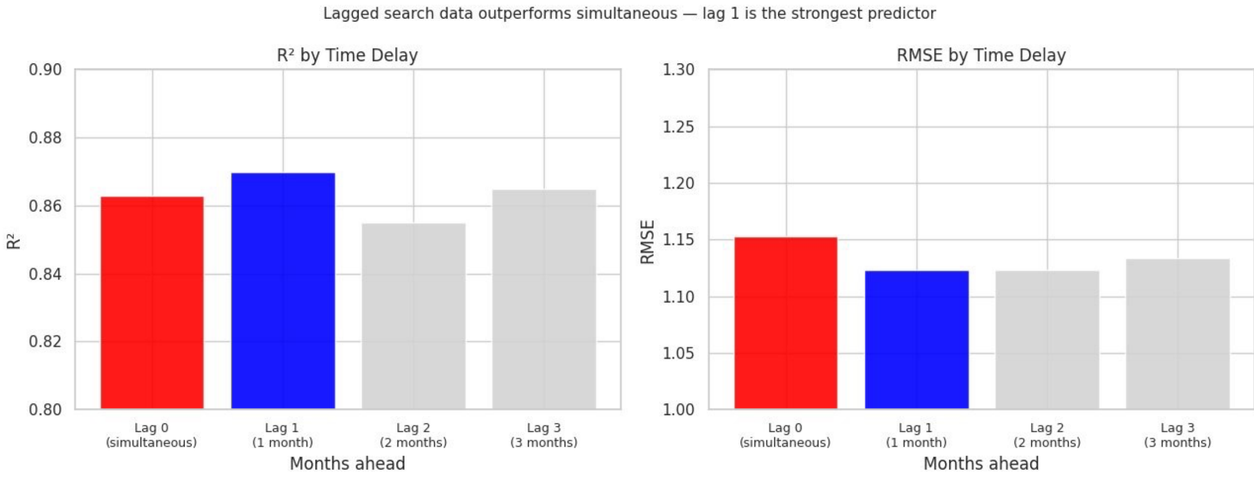

Figure 8. Random Forest performance by time delay (0 = simultaneous).

5.8 Does Adding Economic Variables Help?

Adding macroeconomic variables to the Google Trends features improves R2 from 0.823 to 0.855 and reduces RMSE from 1.236 to 1.123.

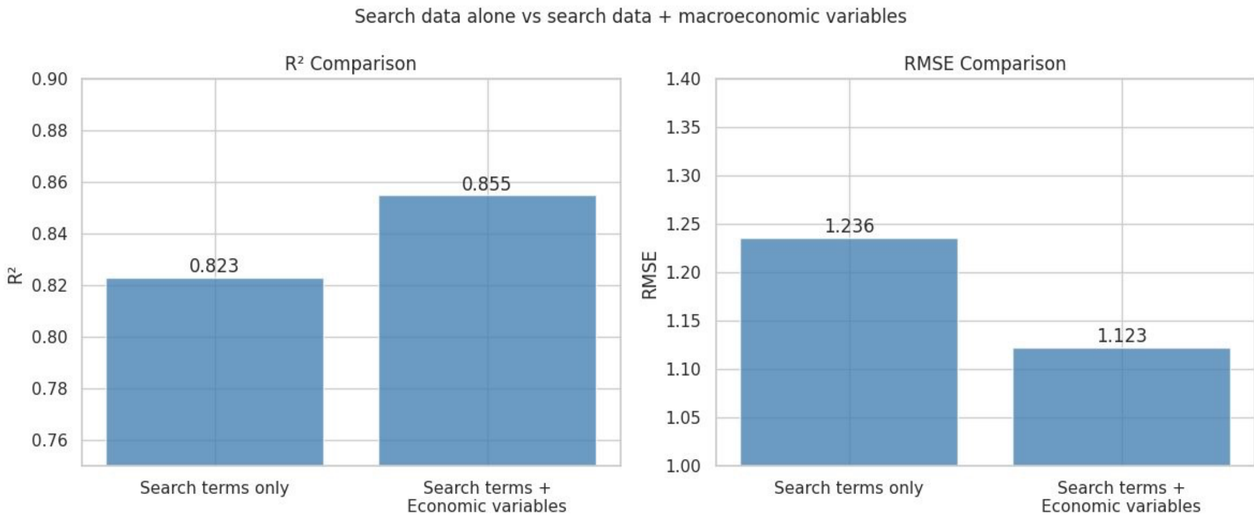

Figure 9. Model performance with search terms only vs search terms and economic variables.

6. Conclusion

The results consistently show that job-related Google search data, particularly işkur, predicts youth unemployment approximately 1 to 2 months in advance. Both the cross-correlation analysis and the Random Forest model point to the same window. The cross correlation analysis identifies lag 2 as the point where the linear relationship between işkur searches and youth unemployment is strongest ($r = 0.54$, $p < 0.05$). The Random Forest model confirms this direction; lag 1 produces the best predictive performance (R2 = 0.874, RMSE = 1.121), slightly outperforming the simultaneous model at lag 0 (R2 = 0.863). Together, these results indicate that a spike in job-related searches consistently precedes a rise in youth unemployment by 1 to 2 months rather than occurring at the same time.

The işkur search term stands out as the strongest predictor across all methods, which makes sense given that it is directly tied to Turkey's public employment agency and is used primarily by people actively looking for work.

One limitation is that Google Trends reports a normalized index rather than absolute search counts, which makes it harder to compare behavior across different time periods. Another limitation is that the dataset only covers 2015 to 2024 so the model may not generalize well to very different economic conditions.

7. Data Sources

- TUİK Labour Force Statistics: data.tuik.gov.tr
- TCMB EVDS: evds2.tcmb.gov.tr
- Google Trends: trends.google.com (collected via pytrends)
- AI Assistance: Claude (Anthropic) was used for code assistance and debugging. Full documentation is available in aiUsage.pdf.